

Pitambar Acharya

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Department of Mathematics

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Applied mathematician and Ph.D. candidate specializing in nonlinear partial differential equations and mathematical modeling, with research interests spanning PDE analysis, computational mathematics, data science, and machine learning. Experienced educator with a broad teaching portfolio across Precalculus, Statistics, Probability, Calculus, Linear Algebra, and other foundational undergraduate mathematics courses.

RESEARCH INTERESTS

Partial Differential Equations • Mathematical Biology & Chemotaxis Models • Data Science • Machine Learning • Scientific Computing

EDUCATION

*Aug 2022 – Expected
Spring 2027*

PhD in Applied and Computational Mathematics

University of Alabama at Birmingham, Birmingham, AL

Dissertation: Global Existence of Solutions on the Quarter Plane for a Chemotaxis Model with Logistic Growth and Logarithmic Sensitivity

Advisor: [Prof. Yanni Zeng](#)

Aug 2025

Master of Science in Mathematics

University of Alabama at Birmingham, Birmingham, AL

Jan 2020 – Jul 2022

Master's in Applied Mathematics and Statistics (Biostatistics Track)

Florida Atlantic University, Boca Raton, FL

2012 – 2014

Master of Science in Pure Mathematics

Tribhuvan University, Kirtipur, Kathmandu, Nepal

2010 – 2012

Master of Science in Physics

Tribhuvan University, Kirtipur, Kathmandu, Nepal

2007 – 2010

Bachelor of Science in Physics (Minor: Mathematics and Statistics)

Trichandra College, Ghantaghar, Kathmandu, Nepal

TEACHING EXPERIENCE

Aug 2022 – Present

Instructor of Record

University of Alabama at Birmingham, Birmingham, AL

- Precalculus: Algebra/Trigonometry (MA 107 QL) – Summer 2026
- Calculus I (MA 125/125L) – Spring 2026
- Precalculus: Trigonometry (MA 106) – Fall 2026, Spring 2025, Fall 2024
- Introduction to Statistics (MA 180) – Fall 2025, Summer 2025
- Precalculus: Algebra (MA 105) – Spring 2024, Fall 2023, Spring 2023
- Intermediate Algebra (MA 102) – Fall 2022

Jan – May 2022

Instructor of Record

Florida Atlantic University, Boca Raton, FL

- Methods of Calculus – Spring 2022

Jan 2020 – Jul 2022

Teaching Assistant

Florida Atlantic University, Boca Raton, FL

- Calculus II (MAC 2312) – Spring 2021, Fall 2020
- Calculus I (MAC 2311) – Spring 2020
- Engineering Mathematics I (MAP 3305) – Summer 2020
- Intermediate & College Algebra: quizzes and midterms – Fall 2021
- Tutor, Engineering Mathematics and Mathematics for Liberal Arts – Summer 2021

Jun 2015 – Dec 2019

Lecturer

Himalaya College of Engineering (Tribhuvan University), Nepal

- Probability, Statistics, Multivariate Statistics and Econometrics for Bachelors in Engineering and Computer Science programs
- Engineering Physics (lectures and labs): Civil, Architecture, Computer, and Electronics Engineering students

PUBLICATIONS

Peer-Reviewed Journal Articles

- [1] K. Acharya and P. Acharya. “Adaptive Filtering via Canonical Systems with Time-Varying Hamiltonians.” *International Journal of Advances in Engineering Sciences and Applied Mathematics* (Springer Nature), 18, 20–30 (2026). DOI: [10.1007/s12572-026-00421-5](https://doi.org/10.1007/s12572-026-00421-5)
- [2] E. P. Poudel, P. Acharya, J. Kafle, and S. Khadka. “On One-Dimensional Advection–Diffusion Equation with Variable Diffusivity.” *Applied Mathematics: A Journal of Chinese Universities* (Springer Nature), 41, 181–193 (2026). DOI: [10.1007/s11766-026-5149-9](https://doi.org/10.1007/s11766-026-5149-9)
- [3] D. Dulal, M. Gallagher, C. Navasca, P. Acharya and N. Skowronski. “Leveraging Advanced Machine Learning to Predict Turbulence Dynamics From Temperature Observations at an Experi-

mental Prescribed Fire.” *IEEE Access*, 14, 112481-112498 (2026) [DOI:10.1109/ACCESS.2026.3715743](https://doi.org/10.1109/ACCESS.2026.3715743).

- [4] E. P. Poudel, P. Acharya, J. Kaffle, and S. Khadka. “Modeling and Analysis of Two-Dimensional Contaminant Dispersion in Turbulent Atmospheric Flows.” Submitted to *Electronic Journal of Differential Equations*.
- [5] D. Bastola, P. Acharya, and D. Dulal. “Predicting Anemia Among Under-Five Children in Nepal Using Machine Learning and Deep Learning.” Submitted to *Public Health Nutrition*. Preprint
- [6] K. Acharya, P. Acharya, and D. Bastola. “Random Schrödinger Operators on Data Graphs: A Data Science Perspective on Robust Spectral Embedding.” Manuscript in preparation.

ADDITIONAL RESEARCH AND PROJECTS

- [1] P. Acharya, L. Zeng, and M. Islam. “Comparative Analysis of Data Mining Techniques for Customer Segmentation and Anomaly Detection.” [Graduate final project](#).
- [2] M. Islam, H. Akther Samia, P. Acharya. “Bin Packing Problem with Machine Learning Techniques.” [View Project](#)
- [3] P. Acharya, O. Francis, and M. Islam. “Deep Q-Learning for Optimizing Bin Packing Problem: A Deep Reinforcement Learning Approach.”

PRESENTATIONS

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|-----------------|---|
| <i>Aug 2025</i> | “Learning the Solution Operator of the Heat Equation.”
<i>Summer School on Scientific Machine Learning, Brin Mathematics Research Center, University of Maryland, College Park</i> |
| <i>Feb 2025</i> | “On One-Dimensional Advection–Diffusion Equation with Variable Diffusivity.”
<i>MAA–FTYCMA Joint Conference, Embry-Riddle Aeronautical University, Daytona Beach</i> |
| <i>May 2022</i> | “Bayesian Model Averaging: An Overview.”
<i>MS Presentation, Florida Atlantic University, Boca Raton, FL</i> |

CONFERENCES & WORKSHOPS ATTENDED

Selected Conferences (recent)

- Fourth Annual Joint Alabama–Florida Conference on Differential Equations, Dynamical Systems & Applications (JAF DEDS) – FIU, May 2026
- 37th Annual UA System Applied Mathematics Meeting – UAH, Nov 2025

- 3rd UNCG Virtual PDE Conference – UNC Greensboro, Oct 2025
- Ohio River Analysis Meeting (ORAM 14) – University of Cincinnati, Mar 2025
- Southeastern Analysis Meeting (SEAM 41) – University of South Florida, Mar 2025
- Modeling and Simulations in Fluids – Brown University (ICERM), Sep 2024
- Joint Mathematics Meetings (JMM) – Boston, MA, Jan 2023

Summer Schools & Workshops

- Summer School: Scientific Machine Learning – Brin MRC, University of Maryland, Aug 2025
- Texas Summer School in Mathematical Physics – Texas A&M University, Jun 2025
- Texas Summer School in Mathematical Physics – Texas State University, Jul 2023
- CIMPA Summer School: Data Visualization, Modeling and Mathematical Tools – Tribhuvan University, Nepal, May 2023 (*Organizing Committee Member*)

LEADERSHIP & SERVICE

<i>Aug 2024 – Present</i>	Secretary – SIAM Student Chapter, UAB
<i>Aug 2022 – Present</i>	Executive Member – Nepalese Students Association, UAB
<i>Jul 2019</i>	Deputy Leader – International Mathematics Olympiad (IMO), Bath, United Kingdom
<i>Jul 2017</i>	Deputy Leader – International Mathematics Olympiad (IMO), Rio de Janeiro, Brazil
<i>Feb 2017 – Dec 2019</i>	Vice President – Mathematical Association of Nepal (MAN) • Trained high school students for the National Mathematics Competition • Conducted examinations and prepared students for IMO • Organized national and international conferences
<i>Aug 2020 – 2021</i>	Vice President – Nepalese Students Association, FAU

VOLUNTEER WORK

- Alabama Statewide High School Mathematics Contest – Feb 2026
- Math Talent Search Grader – Oct 2025
- Alabama Statewide Math Ciphering Contest – Apr 2025
- High School Science Exhibition Judge, Birmingham, AL – Jul 2023
- Kangaroo Mathematics Competition, UAB – Mar 2023
- Math Day Proctoring and Grading, FAU – Jan 2021
- High School Science Olympiad, FAU – Feb 2020
- Blood Donation Program, Kanti Children's Hospital, Kathmandu, Nepal
- Donated stationery and sports materials for cancer-affected children, Kanti Hospital, through

AWARDS & HONORS

<i>Apr 2025</i>	James Robert Ward, Jr. Memorial Endowed Award (\$1,500) for excellence in Graduate Studies in Differential Equations / Mathematical Biology – UAB
<i>Mar 2025</i>	Travel Award (\$563.61) – Ohio River Analysis Meeting, University of Cincinnati
<i>Jun–Jul 2024</i>	Professional Development Stipend (\$1,000) – Blazer Core Curriculum, UAB
<i>Jul 2023</i>	Travel Award (\$1,066) – Texas Summer School in Mathematical Physics, Texas State University
<i>2022 – Present</i>	Graduate Scholarship / Teaching Assistantship – UAB
<i>2020 – 2022</i>	Graduate Scholarship / Teaching Assistantship – FAU
<i>2012 – 2014</i>	MS Fellowship, Central Department of Mathematics, Tribhuvan University
<i>2010 – 2012</i>	MS Fellowship, Central Department of Physics, Tribhuvan University

TECHNICAL SKILLS

Programming Languages	Python, R, MATLAB, SQL, Minitab
Python Libraries	Pandas, NumPy, Matplotlib, SciPy, Scikit-learn, TensorFlow, Keras
Deep Learning	PyTorch (building and training neural networks)
Tools & Platforms	Git, Linux, Jupyter Notebook, AWS, GCP
Statistical Software	R (ggplot2), Minitab

PROFESSIONAL MEMBERSHIPS

- American Mathematical Society (AMS)
- Society for Industrial and Applied Mathematics (SIAM)
- Nepal Mathematical Society (NMS)
- Nepal Physical Society (NPS)
- Association of Nepalese Mathematicians in America (ANMA)

PROFESSIONAL DEVELOPMENT

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- CTL LIFT Teaching Certificate – [Center for Teaching and Learning, UAB](#)
 - Developing Great Presentations – UAB Learning and Development, May 2025
 - Flex/Online Teaching Orientation – Centre for Online and Continuing Education (COCE), FAU
 - Teaching Assistant Basics for Online Courses – COCE, FAU
 - Inquiry-Based Learning Workshop – Department of Mathematical Sciences, FAU
 - CRLA Certification Levels 1 & 2 – Mathematics Learning Center, FAU
 - 14th Annual iLead Conference – Student Union Hall, FAU

LANGUAGES

- English (*professional proficiency*)
- Nepali (*native*)
- Hindi (*proficient*)

REFERENCES

Dr. Yanni Zeng

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University of Alabama at Birmingham
ynzeng@uab.edu (205) 934-2154

Dr. Hongwei Long

Professor
Florida Atlantic University
hlong@fau.edu (561) 297-0810

Dr. Yulia Karpeshina

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University of Alabama at Birmingham
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Dr. Elena Kravchuk

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